

Duality

JEB104 Microeconomics I
Lenka Št'astná

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Duality

Utility Maximization
Expenditure Minimization
Relationships

Indirect Utility Function

Indirect Utility Function

Maximization of utility:

$$\begin{aligned} & \max_{x_1, \dots, x_n} u(x_1, \dots, x_n) \\ \text{s.t. } & \text{(i) } \sum_{i=1}^n p_i x_i \leq m \\ & \text{(ii) } x_i \geq 0, \quad i = 1, \dots, n \end{aligned}$$

Solution: Walrasian/Marshallian demand function for good i $x_i^* = x_i(p, m)$

Indirect utility function:

$$v(p, m) = u(x_1^*, x_2^*, \dots, x_n^*) = u(x_1(p, m), x_2(p, m), \dots, x_n(p, m))$$

= maximized value of $u(x)$ as a function of p and m

Previously

- Utility maximization and expenditure minimization

Today

- The relationship between the two

It's called duality because there are two "dual" problems:

1. Utility max: With given m , consumer tries to reach the highest utility
2. Expenditure min: to reach a given minimum u , consumer tries to spend the lowest budget

You already know that solving utility maximization problems results in Walrasian/Marshallian demand.

Simply plug in the Marshallian demands into the utility function instead of each particular x_i , utility function becomes the function of prices and income then.

Indirect utility function expresses the highest possible utility that can be reached given p and m .

Example: $u(x_1, x_2) = x_1 x_2$

- We know that $x_1 = \frac{m}{2p_1}, x_2 = \frac{m}{2p_2}$
- Hence

$$v(p, m) = u(x(p, m)) = \frac{m}{2p_1} \frac{m}{2p_2} = \frac{m^2}{4p_1 p_2}$$

Numbered example:

- $p_1 = 8, p_2 = 2, m = 80$
- Therefore

$$v(p, m) = \frac{80^2}{4 \cdot 8 \cdot 2} = \frac{6400}{64} = 100$$

- If we want to check the result, we could compute optimal demands: $x_1^*(8, 2, 80) = 5, x_2^*(8, 2, 80) = 20$, and also utility $u(x_1^*, x_2^*) = 5 \cdot 20 = 100$

Lagrange multiplier $\lambda^* = \frac{MU_i}{p_i}$ - how much more utility you can get when adding one additional dollar. Hence, it could be expressed as follows:

$$\lambda^* = \frac{\partial v(p, m)}{\partial m}$$

Prove this for our example above:

$$\begin{aligned} \lambda^* &= \frac{MU_i}{p_i} = \frac{x_2^*}{p_1} = \frac{m}{2p_1 p_2} \\ \lambda^* &= \frac{\partial v(p, m)}{\partial m} = 2 \frac{m}{4p_1 p_2} = \frac{m}{2p_1 p_2} \end{aligned}$$

Indirect Utility Function

If $u(x)$ is a continuous utility function representing monotonic preferences, the indirect utility function $v(p, m)$ is:

- 1 Non-increasing in prices p : $\frac{\partial v(p, m)}{\partial p} \leq 0$
- 2 Increasing in income m
- 3 Homogenous of degree zero in prices p and income m :
 $v(\alpha p, \alpha m) = v(p, m), \quad \alpha > 0$
- 4 Quasiconvex in prices p and income m :
 $v(kp' + (1-k)p'', km' + (1-k)m'') \leq \max\{v(p', m'), v(p'', m'')\}$
- 5 Continuous in prices p and income m

Expenditure Minimization

Minimization of expenditures:

$$\min_{x_1, \dots, x_n} \sum_{i=1}^n p_i x_i$$

s.t. (i) $u(x) \geq \bar{u}$

(ii) $x_i \geq 0, \quad i = 1, \dots, n$

Solution: Hicksian demand function $x_i^* = h_i(p, u)$

Expenditure function:

$$e(p, u) = \sum_{i=1}^n p_i x_i^* = \sum_{i=1}^n p_i h_i(p, u)$$

= minimized spending as a function of p and u

Non-increasing in p

- If price of a good increases: if not purchased then optimum does not change, optimal utility stays the same (equality)
- If price of good increases (and if purchased), then new bundle purchased.
- Old bundle not available any more. New bundle available under old prices, utility in new situation has to be lower (strictly convex preferences).

Increasing in m

- higher income = new optimum with x , which was not available under old income. (Old bundle available, hence utility has to increase).

$$v(\alpha p, \alpha m) = v(p, m)$$

- Budget constraint does not change, relative prices same, optimal bundle remains the same, so maximized level of utility does not change.

Quasiconvex in prices

- See graphical proof at the end for those who are interested

You already know that solving expenditure minimization problems results in Hicksian demand.

Simply plug in Hicksian demands into the budget constraint instead of each particular x_i , so expenditure function as a function of prices and utility expresses the minimal level of spending we need to spend under given prices p in order to reach a required level of utility u .

Example: $u(x_1, x_2) = x_1 x_2$

- We know that $h_1 = \sqrt{\frac{p_2 u}{p_1}}, h_2 = \sqrt{\frac{p_1 u}{p_2}}$
- Hence

$$e(p, u) = \sum_{i=1}^n p_i h_i(p, u) = p_1 \sqrt{\frac{p_2 u}{p_1}} + p_2 \sqrt{\frac{p_1 u}{p_2}} = 2\sqrt{p_1 p_2 u}$$

Numbered example:

- $p_1 = 8, p_2 = 2$, required level of utility $u = 100$
- Therefore

$$e(p, u) = 2\sqrt{8 \cdot 2 \cdot 100} = 2\sqrt{1600} = 80$$

- If we want to check the result, we could compute optimal demands:
 $h_1(8, 2, 100) = 5, h_2(8, 2, 100) = 20$, and also spending $8 \cdot 5 + 2 \cdot 20 = 80$

Lagrange multiplier $\lambda^* = \frac{p_i}{MU_i}$ - how costly would it be to increase utility by one util.

$$\lambda^* = \frac{\partial e(p, u)}{\partial u}$$

Prove this for our example above (note that you plug $h_2(p_1, p_2, u)$ instead of x_2):

$$\lambda^* = \frac{p_i}{MU_i} = \frac{p_1}{x_2} = \frac{p_1}{\sqrt{\frac{p_1 u}{p_2}}} = \frac{p_1 \sqrt{p_2}}{\sqrt{p_1 u}} = \frac{\sqrt{p_1 p_2}}{\sqrt{u}}$$

$$\lambda^* = \frac{\partial e(p, u)}{\partial u} = \frac{1}{2} \frac{2\sqrt{p_1 p_2}}{\sqrt{u}}$$

Expenditure Function $e(p, u)$

If $u(x)$ is a continuous utility function representing locally non-satiated preferences, the expenditure function $e(p, u)$ is:

- ➊ Increasing in utility u : $\frac{\partial e(p, u)}{\partial u} > 0$
- ➋ Non-decreasing in prices p : $\frac{\partial e(p, u)}{\partial p} \geq 0$
- ➌ Linearly homogeneous in prices p : $e(\alpha p, u) = \alpha \cdot e(p, u)$, $\alpha > 0$
- ➍ Concave in prices p : $\frac{\partial^2 e(p, u)}{\partial p^2} \leq 0$
- ➎ Continuous in prices p and utility u

Reciprocity of Expenditure Minimization and Utility Maximization

UMP (Utility Maximization Problem):

- bundles in the budget set $B = \{x \in \mathbb{R}_{+,0}^n : \sum_{i=1}^n p_i x_i \leq m\}$
- choose the one touching the highest indifference curve

EMP (Expenditure Minimization Problem):

- bundles in the upper contour set of x , when $u(y) \geq u(x) = \bar{u}$:
 $P_x = \{y \in \mathbb{R}_{+,0}^n : y \succeq x\}$
- choose the one touching the lowest isocost line

Given prices p , consumer has $\bar{m}$:

- max utility $\rightarrow x^*$, utility $u(x^*)$

Given prices p , consumer aims to reach $\bar{u}$:

- min expenditure $\rightarrow y^*$, spending $\sum_{i=1}^n p_i y_i^*$

$$x^* = y^* \iff u(x^*) = \bar{u} \iff \bar{m} = \sum_{i=1}^n p_i y_i^*$$

Increasing in utility u

- Higher utility requires higher spending if in optimum. (Optimum lies on higher isocost line)
- Reaching higher utility without higher spending would mean that we are not in optimum.

Non-decreasing in prices p

- If not purchased in the optimum, the change in price of good k does not affect expenditures = no change.
- If purchased and its price increases, new bundle is chosen. But this bundle is not optimal under old prices p' (bundle is not minimizing spending under old prices).
- Spending on the original bundle x' : $e(p', u) = p'x'$
- Price increases $p'' > p'$, spending on the new bundle x'' : $e(p'', u) = p''x''$
- New bundle is more costly under p'' than under p' , just because $p'' > p'$, so $p''x'' > p'x''$
- Under prices p' , it was the original bundle x' that minimized spending to reach a utility level u , so $p'x'' > p'x'$
- Putting previous two relations together gives: $p''x'' > p'x'$, so spending increases with increase of price (given that the good is purchased).

Linearly homogeneous in prices p

- Problem does not change, relative prices remain the same, so multiplication by α linearly projects to spending.

Concave in prices p

- Second-order partial derivatives
- Suppose price of good k increases (for example by 10%), then if the original optimal bundle is purchased even after the price change, spending on good x_k increases linearly with the price change (also by 10 %). Second-order partial derivative would be zero.
- But consumer adjusts consumption bundle because of changes in relative prices and substitutes more expensive good with relatively cheaper goods (equivalent to substitution effect). So with increase of its price, consumer demand less of a good whose price increased (notice that we want to reach a given utility level), so it is exactly substitution effect and movement along Hicksian demand.
- Hence, growth rate of spending is lower than growth rate of price.

The UMP and EMP are “dual” or “reciprocal”.

The UMP and EMP result in the same demanded bundles, if income available in the UMP $\bar{m}$ is the same as a minimized level of spending necessary to reach a utility level of $\bar{u}$ in the EMP. At the same time, the maximized level of utility that can be reached in the UMP is exactly the same as the required level of utility in the EMP $\bar{u}$.

Expenditure Function and Hicksian Demand

Hicksian demands can be extracted from the expenditure function

Shepard's lemma:

$$\frac{\partial e(p, u)}{\partial p_i} = h_i(p, u) \quad \forall p_i, i = 1, \dots, n$$

► Proof

Relationship between the expenditure function and the Hicksian demand

Hicksian demand is the derivative of the expenditure function

Example: $u(x_1, x_2) = x_1 x_2$

- We know that $e(p, u) = 2\sqrt{p_1 p_2 u} = 2(p_1 p_2 u)^{\frac{1}{2}}$
- Hence $h_i = \frac{\partial e(p, u)}{\partial p_i}$

$$h_1 = 2 \cdot \frac{1}{2} p_2 u (p_1 p_2 u)^{-\frac{1}{2}} = \sqrt{\frac{p_2 u}{p_1}}$$

$$h_2 = 2 \cdot \frac{1}{2} p_1 u (p_1 p_2 u)^{-\frac{1}{2}} = \sqrt{\frac{p_1 u}{p_2}}$$

Proof optional (you may take a look at the end of the presentation).

Indirect Utility Function and Walrasian Demand

Walrasian demands can be extracted from the indirect utility function

Roy's identity:

$$x_i(p, m) = - \frac{\frac{\partial v(p, m)}{\partial p_i}}{\frac{\partial v(p, m)}{\partial m}} \quad \forall p_i, i = 1, \dots, n$$

► Proof

Relationship between indirect utility function and Walrasian demand

Intuitive explanation

Rate of change of maximized utility with respect to price ($\frac{\partial v(p, m)}{\partial p_i}$) is equal to product of:

- Rate of change of real income with this price (tj. $-x_i$ - the number of units that have changed, so our original demand for good whose price changes)
- Rate the maximized utility changes with change of income (tj. $\lambda_i = \frac{\partial v(p, m)}{\partial m}$)

Example: $u(x_1, x_2) = x_1 x_2$

- We know that $v(p, m) = \frac{m^2}{4p_1 p_2}$
- Hence $x_i(p, m) = - \frac{\frac{\partial v(p, m)}{\partial p_i}}{\frac{\partial v(p, m)}{\partial m}}$

$$x_1 = - \frac{\frac{m^2}{4p_2} p_1^{-2}}{\frac{2m}{4p_1 p_2}} = \frac{m}{2p_1}$$

$$x_2 = - \frac{\frac{m^2}{4p_1} p_2^{-2}}{\frac{2m}{4p_1 p_2}} = \frac{m}{2p_2}$$

Proof optional (you may take a look at the end of the presentation).

Indirect Utility Function, Expenditure Function and Demands

$v(p, m)$ and $e(p, u)$ are inverse functions:

$$\bar{u} = v(p, e(p, \bar{u}))$$

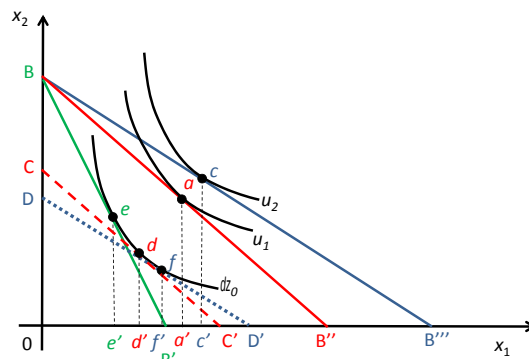
$$\bar{m} = e(p, v(p, \bar{m}))$$

With $v(p, m)$ ($e(p, u)$), we can transform $h(p, u)$ into $x(p, m)$ (vice versa).

$$x(p, \bar{m}) = h(p, v(p, \bar{m}))$$

$$h(p, \bar{u}) = x(p, e(p, \bar{u}))$$

Derivation of Hicksian and Walrasian Demand Curves



- ① Point e. Budget BB' , $x_1 = e'$ both from UMP and EMP (u_0 and m_0).
- ② $p_1 \downarrow$. UMP: income m_0 , budget BB'' , $x_1 = a'$. EMP: utility u_0 , $x_1 = d'$.
- ③ $p_1 \downarrow$ again. UMP: $x_1 = c'$. EMP: $x_1 = f'$.

Inverse functions: $y = f(x) \Leftrightarrow x = f^{-1}(y) = g(y)$, so $f(\cdot)$ and $g(\cdot)$ are inverse functions: example $y = x^2$ and $x = \sqrt{y}$

$$e^{-1}(\cdot) = v(\cdot) \text{ and } v^{-1}(\cdot) = e(\cdot)$$

Example: $u(x_1, x_2) = x_1 \cdot x_2$

- We know that $v(p, m) = \frac{m^2}{4p_1p_2}$, $e(p, u) = 2\sqrt{p_1p_2u}$
- Hence, the following must hold:

$$v(p, e(p, u)) = \frac{(2\sqrt{p_1p_2u})^2}{4p_1p_2} = u \quad e(p, v(p, m)) = 2\sqrt{p_1p_2 \left(\frac{m^2}{4p_1p_2} \right)} = m$$

and we can easily get $e(p, u)$ once we know $v(p, m)$ and vice versa:

$$v(p, m) = \frac{m^2}{4p_1p_2} = u \Rightarrow m = \sqrt{u \cdot 4p_1p_2} = 2\sqrt{p_1p_2u} = e(p, u)$$

$$e(p, u) = 2\sqrt{p_1p_2u} = m \Rightarrow u = \left(\frac{m}{2\sqrt{p_1p_2}} \right)^2 = \frac{m^2}{4p_1p_2} = v(p, m)$$

- We also know that $x_1 = \frac{m}{2p_1}$, $h_1 = \sqrt{\frac{p_2u}{p_1}}$, hence

$$h(p, v(p, m)) = \sqrt{\frac{p_2 \frac{m^2}{4p_1p_2}}{p_1}} = \frac{m}{2p_1} = x(p, m)$$

$$x(p, e(p, u)) = \frac{2\sqrt{p_1p_2u}}{2p_1} = \sqrt{\frac{p_2u}{p_1}} = h(p, u)$$

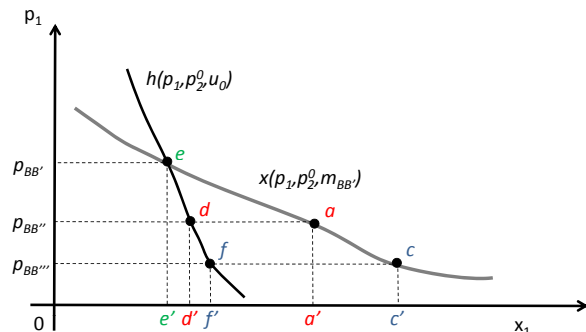
and analogically for x_2 and h_2

1. **Point e** – starting point. Given budget line BB' , $x_1 = e'$ maximizes utility of a consumer (his utility is u_0). Given utility level u_0 and relative prices, $x_1 = e'$ minimizes expenditures of a consumer to reach u_0 (these expenditures are m_0). So point e lies on both Walrasian and Hicksian demand.
2. P_1 **decreases**. With income m_0 new budget line BB'' can be reached. Utility maximizing consumer demands $x_1 = a'$. Consumer with goal of reaching utility u_0 minimizes expenditures if he demands $x_1 = d'$.
3. P_1 **decreases again**. With income m_0 new budget line BB''' can be reached. Utility maximizing consumer demands $x_1 = c'$. Consumer aiming at reaching utility u_0 minimizes expenditures if he demands $x_1 = f'$.

Points e, a and c are points on Walrasian demand, while points e, d, f are points on Hicksian demand.

Notice that the change from e' do d' is the SE and the change from d' do a' is the IE.

Hicksian and Walrasian Demand Curves



- Cross when required utility level u_0 in EMP is maximized level of utility for given income $m_{BB'} = m_0$ in UMP
- Change in price $\rightarrow$ utility does/doesn't change (Walrasian/Hicksian)
- Normal good: Walrasian demand always flatter than Hicksian demand
- Inferior good: Hicksian demand always flatter than Walrasian demand

Relationship between Walrasian and Hicksian Demand Functions

From $h(p, \bar{u}) = x(p, e(p, \bar{u}))$:

$$\frac{\partial h_j(p, \bar{u})}{\partial p_i} = \frac{\partial x_j(p, e(p, \bar{u}))}{\partial p_i} + \underbrace{\frac{\partial x_j(p, e(p, \bar{u}))}{\partial e(p, \bar{u})} \cdot \frac{\partial e(p, \bar{u})}{\partial p_i}}_{h_i(p, \bar{u})}$$

Since $e(p, \bar{u}) = \bar{m}$ and $h_i(p, \bar{u}) = x_i(p, \underbrace{e(p, \bar{u})}_{\bar{m}})$

$$\frac{\partial h_j(p, \bar{u})}{\partial p_i} = \frac{\partial x_j(p, \bar{m})}{\partial p_i} + \frac{\partial x_j(p, \bar{m})}{\partial \bar{m}} \cdot x_i(p, \bar{m})$$

Slutsky equation:

$$\frac{\partial x_i(p, m)}{\partial p_j} = \frac{\partial h_i(p, u)}{\partial p_j} + (-x_j(p, m)) \cdot \frac{\partial x_i(p, m)}{\partial m}, \quad i, j \in \{1, 2, \dots, n\}$$

Relationship between slopes of Hicksian and Walrasian demand

The figure shows the Hicksian (compensated) and the Walrasian (uncompensated) demands derived from the previous slide. The demand are not the same.

Walrasian flatter than Hicksian for normal goods

- The difference between the two curves is the income effect
- In the case of normal good, with decrease of its price (our real income increases) we demand more of the good (and reach higher level of utility).

Walrasian steeper than Hicksian for inferior goods

- Check the previous graph: $a' < d'$ is possible only if x_1 is inferior
- In the case of inferior good, with decrease of its price (our real income increases) we demand less of the good.

Previous slides was a graphical demonstration, now formally.

There is no need to remember this derivation. There **is** a need to remember the **Slutsky equation**.

Notice that p_j in the Slutsky equation can also be p_i (does not hold only for a cross-price change, but also for own price change).

Slutsky Equation

Slutsky equation for own-price change:

$$\underbrace{\frac{\partial x_i(p, m)}{\partial p_i}}_{\text{Total price effect}} = \underbrace{\frac{\partial h_i(p, u)}{\partial p_i}}_{\text{Substitution effect}} + \underbrace{(-x_i(p, m)) \cdot \frac{\partial x_i(p, m)}{\partial m}}_{\text{Income effect}}$$

Slope of Walrasian demand Slope of Hicksian demand

(slope is negative, demand not inverse in Slutsky equation)

$$\text{Normal good: } \frac{\partial x_i(p, m)}{\partial p_i} = \frac{\partial h_i(p, u)}{\partial p_i} + \underbrace{(-x_i(p, m)) \frac{\partial x_i(p, m)}{\partial m}}_{<0} \Rightarrow \frac{\partial x_i(p, m)}{\partial p_i} < \frac{\partial h_i(p, u)}{\partial p_i}$$

Walrasian demand flatter than Hicksian demand

$$\text{Inferior good: } \frac{\partial x_i(p, m)}{\partial p_i} = \frac{\partial h_i(p, u)}{\partial p_i} + \underbrace{(-x_i(p, m)) \frac{\partial x_i(p, m)}{\partial m}}_{>0} \Rightarrow \frac{\partial x_i(p, m)}{\partial p_i} > \frac{\partial h_i(p, u)}{\partial p_i}$$

Hicksian demand flatter than Walrasian demand

Slutsky Equation: Implications

Normal good is always ordinary good:

$$\underbrace{\frac{\partial x_i(p, m)}{\partial p_i}}_{<0} = \underbrace{\frac{\partial h_i(p, u)}{\partial p_i}}_{<0} + \underbrace{(-x_i(p, m)) \frac{\partial x_i(p, m)}{\partial m}}_{<0}$$

Giffen good must be inferior good:

$$\underbrace{\frac{\partial x_i(p, m)}{\partial p_i}}_{>0} = \underbrace{\frac{\partial h_i(p, u)}{\partial p_i}}_{<0} + \underbrace{(-x_i(p, m)) \frac{\partial x_i(p, m)}{\partial m}}_{>0}$$

Inferior good may be ordinary or Giffen:

$$\left| \underbrace{\frac{\partial h_i(p, u)}{\partial p_i}}_{<0} \right| \leq \left| \underbrace{(-x_i(p, m)) \frac{\partial x_i(p, m)}{\partial m}}_{>0} \right|$$

SE: $\frac{\partial h_i(p, u)}{\partial p_i}$ is equivalent to $\frac{\Delta x_i^*}{\Delta p_i} \big|_{u(x)=u^0}$ covered in previous lecture

IE:

- Change in real income $(-x_j(p, m))$ equivalent to $\underbrace{\frac{\Delta m_{real}}{\Delta p_i}}_{<0}$ because

$$\Delta m_{real} = \Delta p_i \cdot x_i$$

- Sensitivity to income change $\frac{\partial x_i(p, m)}{\partial m}$ equivalent to $\frac{\Delta D_i(p, m)}{\Delta m}$

Be cautious when comparing slopes: Slopes for downward-sloping demands are negative and in graphs, we usually draw inverse demands (derivatives in Slutsky equations are slopes of demands drawn using correct mathematical approach and not economic approach which is to draw inverse demands).

Most of these things we should already know based on the previous lectures, but the Slutsky equation will show us right away and formally.

Normal good will always have a downward-sloping demand (cannot be Giffen).

Giffen good has to be inferior.

But inferior good does not have to be Giffen.

Slutsky Equation: Cross-Price Effects

Net complements

Normal good is also gross complement:

$$\underbrace{\frac{\partial x_i(p, m)}{\partial p_j}}_{<0} = \underbrace{\frac{\partial h_i(p, u)}{\partial p_j}}_{<0} + \underbrace{(-x_j(p, m)) \cdot \frac{\partial x_i(p, m)}{\partial m}}_{<0}$$

Net substitutes

Inferior good is also gross substitute:

$$\underbrace{\frac{\partial x_i(p, m)}{\partial p_j}}_{>0} = \underbrace{\frac{\partial h_i(p, u)}{\partial p_j}}_{>0} + \underbrace{(-x_j(p, m)) \cdot \frac{\partial x_i(p, m)}{\partial m}}_{>0}$$

We've also seen these before, now they can all be deduced from Slutsky equation.

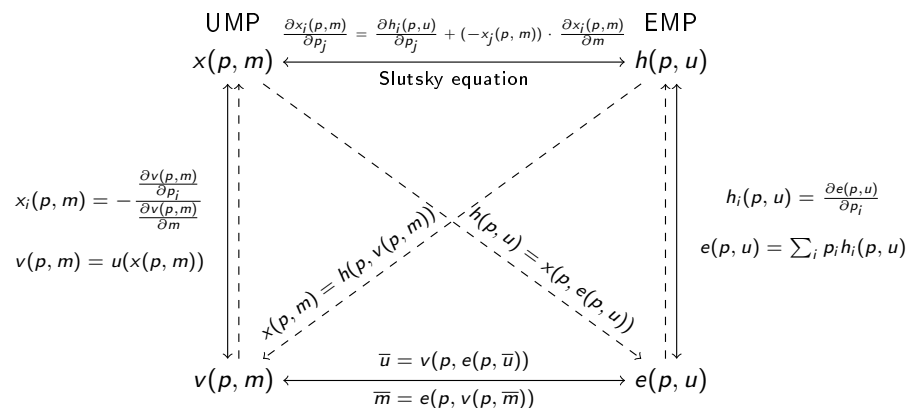
Net substitutes/complements determined by a sign of $\frac{\partial h_i(p, u)}{\partial p_j}$ (substitutes > 0 , complements < 0).

If net complement and inferior, it can be gross complement or substitute.

If net substitute and normal, it can be gross complement or substitute.

Duality: Summary

A summary of duality - one can quickly glance and remind the relationships



Shepard's Lemma: Proof

$$\frac{\partial e(p, u)}{\partial p_i} = h_i(p, u) \quad \forall p_i, i = 1, \dots, n$$

Proof:

$$e(p, u) = \sum_{j=1}^n p_j h_j(p, u)$$

$$\frac{\partial e(p, u)}{\partial p_i} = h_i(p, u) + \sum_{j=1}^n p_j \frac{\partial h_j(p, u)}{\partial p_i}$$

From EMP: $\frac{\lambda^* \partial u(x^*)}{\partial x_j} = p_j$

Hence,

$$\frac{\partial e(p, u)}{\partial p_i} = h_i(p, u) + \underbrace{\lambda^* \sum_{j=1}^n \frac{\partial u(x^*)}{\partial x_j} \frac{\partial h_j(p, u)}{\partial p_i}}_0$$

$$u(x^*) = u(h(p, u)) = \bar{u} \rightarrow \frac{\partial u(x^*)}{\partial p_i} = \frac{\partial u(x^*)}{\partial x_j} \frac{\partial h_j(p, u)}{\partial p_i} = 0$$

[◀ Go back](#)

Proof optional (only for those that are interested)

Roy's Identity: Proof

$$x_i(p, m) = - \frac{\frac{\partial v(p, m)}{\partial p_i}}{\frac{\partial v(p, m)}{\partial m}} \quad \forall p_i, i = 1, \dots, n$$

Proof:

$$v(p, m) = u(x_1(p, m), x_2(p, m), \dots, x_n(p, m))$$

$$\frac{\partial v(p, m)}{\partial p_i} = \sum_{j=1}^n \frac{\partial u(x)}{\partial x_j} \frac{\partial x_j(p, m)}{\partial p_i}$$

From UMP: $\frac{\partial u(x^*)}{\partial x_j} = \lambda^* p_j$ and $\lambda^* = \frac{\partial u(x^*)}{\partial m} = \frac{\partial v(p, m)}{\partial m}$, hence

$$\frac{\partial v(p, m)}{\partial p_i} = \underbrace{\lambda^*}_{\frac{\partial v(p, m)}{\partial m}} \underbrace{\sum_{j=1}^n p_j \frac{\partial x_j(p, m)}{\partial p_i}}_{-x_i(p, m)}$$

From Walras' law: $\sum_{j=1}^n p_j x_j(p, m) = m \rightarrow \text{derivative wrt } p_i \quad x_i(p, m) + \sum_{j=1}^n p_j \frac{\partial x_j(p, m)}{\partial p_i} = 0$

[◀ Go back](#)

Proof optional (only for those that are interested)

Indirect utility function - quasiconvex in p and m

$$v(kp' + (1-k)p'', km' + (1-k)m'') \leq \max \{ v(p', m'), v(p'', m'') \}$$

Graphical proof:

$$(p', m') \rightarrow BC' : p'_1 x_1 + p'_2 x_2 \leq m' \xrightarrow{\text{Budget lines}} x_2 = \frac{m'}{p'_2} - \frac{p'_1}{p'_2} x_1$$

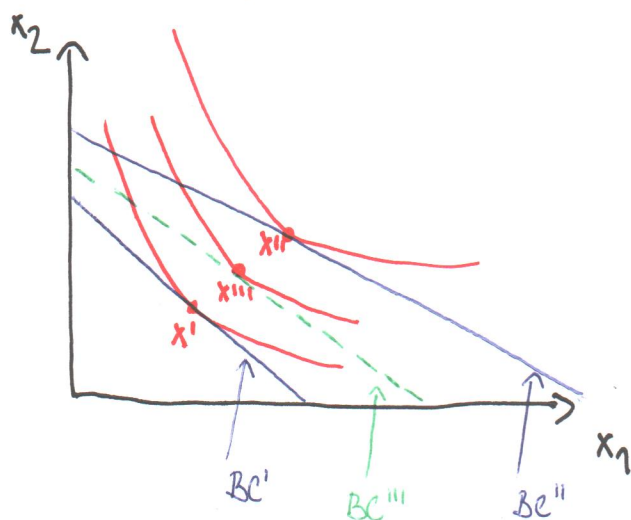
$$(p'', m'') \rightarrow BC'' : p''_1 x_1 + p''_2 x_2 \leq m'' \xrightarrow{\text{Budget lines}} x_2 = \frac{m''}{p''_2} - \frac{p''_1}{p''_2} x_1$$

Suppose :

① $v(p', m') \neq v(p'', m'')$

$\Rightarrow$ maximized utility levels are different, so $x' = x^*(p', m')$ and $x'' = x^*(p'', m'')$ lie on different indifference curves

(let's say $u(x') < u(x'')$)

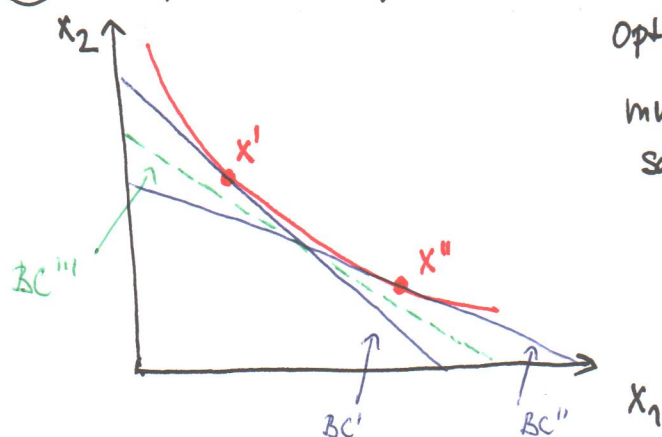


$$BC''' : (kp'_1 + (1-k)p''_1)x_1 + (kp'_2 + (1-k)p''_2)x_2 \leq km' + (1-k)m''$$

$\rightarrow$ budget line will lie somewhere between budget lines of BC' and BC'' depending on parameter k

$\Rightarrow u(x''') < \max \{ u(x'), u(x'') \} \Rightarrow$ equivalent to quasiconvexity of $v(p, m)$

② $v(p', m') = v(p'', m'')$



Optimum lying on budget line of BC''' (green) must be on IC that is lower (or the same) than IC corresponding to utility level $u(x') = u(x'')$

$$\Rightarrow u(x''') \leq \max \{ u(x'), u(x'') \}$$